

# SI 140A-02 Probability & Statistics for EECS, Fall 2026

## Homework 2

Name:

Student ID:

Due on Oct. 4, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.
- You are required to **submit your homework in PDF format generated by LaTeX**, otherwise you will **get no points of this homework**.

**Problem 1 (25%)**

A box contains 5 red balls, 4 blue balls, and 3 green balls. Three balls are drawn sequentially without replacement.

Let  $A = \{\text{the first ball is red}\}$ , and let  $C = \{\text{at least one of the three drawn balls is red}\}$ .

Suppose that the only information revealed after the experiment is that event  $C$  occurred; that is, at least one of the three drawn balls was red.

- (a) Find  $P(C)$ .
- (b) Find  $P(A \cap C)$  and hence calculate  $P(A | C)$ .
- (c) Let  $B$  be the event that the second ball is red. Calculate  $P(B | C)$ .
- (d) Calculate  $P(A \cap B | C)$ .
- (e) Compare  $P(A | C)$  with  $P(A)$ . Are  $A$  and  $C$  independent? Justify your answer using the definition of independence.

**Problem 2 (25%)****Multiple Detection System**

A monitoring system uses four independent sensors to detect whether a machine has experienced a failure.

Let  $A_i$  denote the event that sensor  $i$  raises a false alarm.

The false-alarm probabilities are  $P(A_1) = 0.10$ ,  $P(A_2) = 0.15$ ,  $P(A_3) = 0.20$ ,  $P(A_4) = 0.25$ .

Assume that the four events are mutually independent.

The monitoring system issues a warning if **at least one sensor** raises a false alarm.

- (a) Suppose the monitoring system has issued a warning. Find the probability that **at least two sensors** have raised a false alarm, i.e.,  $P(\text{at least two alarms} \mid A_1 \cup A_2 \cup A_3 \cup A_4)$ .
- (b) Suppose now that no independence assumption is made. Instead, you are given  $P(A_i) = 0.1$ ,  $i = 1, 2, 3, 4$ , and  $P(A_i \cap A_j) = 0.02$  for every  $i \neq j$ , together with  $P(A_i \cap A_j \cap A_k) = 0.005$  for every distinct  $i, j, k$ , while  $P(A_1 \cap A_2 \cap A_3 \cap A_4) = 0.001$ .

Use the inclusion-exclusion principle to calculate  $P(A_1 \cup A_2 \cup A_3 \cup A_4)$ .

**Problem 3 (25%)**

A factory has three production lines,  $L_1$ ,  $L_2$ , and  $L_3$ , which produce 30%, 45%, and 25% of the factory's total products, respectively. The probabilities that a product is defective given that it comes from  $L_1$ ,  $L_2$ , or  $L_3$  are  $P(D | L_1) = 0.02$ ,  $P(D | L_2) = 0.03$ ,  $P(D | L_3) = 0.05$ , where  $D$  denotes the event that a randomly selected product is defective.

- (a) Two products are selected independently from the factory. Find the probability that both products are defective.
- (b) Three products are selected independently from the factory. Find the probability that at least one of the three products is defective.
- (c) Suppose  $n$  products are selected independently from the factory. Find the smallest integer  $n$  such that the probability of obtaining at least one defective product is at least 95%.

**Problem 4 (25%)**

Two dice are rolled independently. For each die, the probability of showing an even number is  $p$ , where  $0 < p < 1$ . The two dice have the same distribution, but no assumption is made that the individual outcomes  $1, 2, \dots, 6$  are equally likely.

Define  $A = \{\text{the sum of the two dice is even}\}$ , and  $B = \{\text{the sum of the two dice is greater than 8}\}$ .

Determine whether there exists a value of  $p$  such that, for every possible distribution of the individual die outcomes satisfying  $P(\text{even}) = p$ , the events  $A$  and  $B$  are independent.

If such a value of  $p$  exists, find all such values and prove that the independence holds for every admissible distribution.

If no such value exists, prove this by constructing, for each candidate value of  $p$ , two admissible distributions that have the same value of  $p$  but give different values of  $P(A \cap B) - P(A)P(B)$ .

Your argument should distinguish carefully between independence that is implied by the given information and independence that occurs only for a particular choice of the underlying distribution.